CF14: Principle of Stationary Action as Temporal Effective Invariant

Complete Formalism

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Abstract

This paper derives the principle of stationary action from the Primitive Bifurcation Law of the Void Dynamics Model (VDM), showing that the familiar variational rule is not an independent postulate but the temporal-path effective invariant generated when the primitive non-discharge law is expressed on an admitted temporal domain. CF 000 proves that same-domain articulation cannot continue indefinitely: once lawful invariant-bearing articulation within a domain is exhausted while discharge remains impossible, orthogonal re-articulation into a new irreducible domain is forced. The universal axiom paper A(-1) sharpens this constitutional picture: void debt begins exactly at the saturation limit, and every later invariant and later axiom is effective rather than primitive. CF 00 then makes the first orthogonal articulation mechanically explicit: the quarter-turn operator is algebraically borne by i, while the half-turn completion/opening condition is measured by π. CF 14 shows that when temporal relation becomes admissible, the same rule reappears as a law on endpoint-fixed admissible paths: continuation proceeds by the minimum admissible accumulated orthogonal articulation required to keep the invariant borne without discharge. This yields the stationarity condition $\delta S_{\mathcal{I}} = 0$, from which the Euler–Lagrange equations follow as the local path-level non-discharge balance law. We then order the familiar variational principles by dependency burden and derive them as realised descendants of the same root law: Maupertuis' principle, Hamilton's principle, thermodynamic-potential extremisation, Onsager's dissipative principle, local field-action stationarity, gauge action stationarity, gravitational action stationarity, Schwinger's quantum action principle, the Feynman path integral, and variational state-space descendants such as Rayleigh–Ritz and time-dependent variational principles. Stationary action is therefore the temporal effective invariant of minimum admissible orthogonal articulation.

Keywords stationary action · least action · effective invariant · effective axiom · primitive bifurcation · orthogonal articulation · void debt · minimum saturation · Euler–Lagrange · Onsager · path integral

1 Introduction

Motivation

The principle of least action sits at the heart of modern physics, appearing in mechanics, thermodynamics, field theory, gravity, and quantum theory. In conventional presentations it is treated as a basic variational postulate. VDM instead requires a stricter discipline: no later law may be primitive if it can be derived from the already-earned invariant spine. The aim of CF 14 is therefore not merely to reinterpret stationary action but to derive it.

Series Placement

CF 14 occupies a precise place in the complete formalism series. CF 000 closes the root invariant spine: admissible origin, two-pole non-discharge, same-axis articulation, saturation, and forced orthogonal re-articulation into 2D. A(-1) states the constitutional law in universal form: there is one primitive law; later invariants and later axioms are effective; void debt begins at the saturation limit; and minimum protection forbids importing more structure than is required to keep the invariant borne. CF 00 then makes the first orthogonal articulation mechanically explicit: the quarter-turn operator is borne by i, the continuous half-turn completion/opening condition is measured by π, and lawful variation/differentiability and the carrier domain are earned. CF 01 subsequently derives the explicit QGT-to-metriplectic split $J \oplus M$. CF 02 closes the contact-geometric to metriplectic thermodynamic bridge and therefore the formal thermodynamic hosting of the dissipative limb. The missing bridge is then the one CF 14 closes: why variational physics exists at all, and why its stationary-action law is the same invariant law seen on an admitted temporal path domain.

Scope and Contribution

This paper operates at two theorem-bearing levels.

(i) It proves the primitive temporal theorem: among admissible endpoint-fixed paths, the selected path carries no avoidable first-order excess in accumulated invariant-bearing orthogonal articulation cost. In this sense, stationary action is the temporal-domain effective invariant of the same minimum-protection rule that already forced the first orthogonal articulation.

(ii) It then orders the known variational principles from least dependencies to most dependencies and derives them as realised descendants of the same law. Maupertuis' principle, Hamilton's principle, thermodynamic extremisation, dissipative Onsager flow, field-action stationarity, gauge action stationarity, Einstein–Hilbert stationarity, Schwinger's quantum action principle, the Feynman path integral, and state-space variational descendants are not independent beginnings; they are richer domain-specific closures of the same primitive rule.

Historical Naming Note

Throughout this paper, "least action" is used only in the historical sense. The formal theorem proved here is stationarity of action, that is, absence of avoidable first-order excess. A selected path may therefore be a minimum, maximum, or saddle with respect to higher-order variations; CF 14 closes only the first-order law.

What This Work Is Not

- This paper is *not* a re-derivation of QGT, Berry curvature, the carrier, or the explicit $J \oplus M$ split. Those are already earned in CF 00/CF 01 and are integrated here only as already-closed descendants.
- It is *not* a full derivation of every later field equation from scratch. It derives the root variational law and shows how each realised principle follows from it once the relevant added structure is admitted.
- It is *not* a runtime or empirical paper. No code or simulations are introduced here.

2 Constitutional Background: Primitive Law, Effective Invariants, and Orthogonal Articulation

One Primitive Law; Later Effective Invariants and Effective Axioms

VDM does not begin from many primitive laws. It begins from one primitive invariant: unresolved two-pole opposition borne internally by one admissible origin-condition. Because that invariant cannot discharge into either isolated pole, it must continue articulating itself wherever lawful articulation remains possible. A(-1) makes the constitutional consequence explicit: later invariants are *effective invariants*, and later laws are *effective axioms*. They are not new primitive beginnings; they are domain-proper expressions of the same primitive invariant once a new class/domain has been forced and minimally stabilised.

Dependence Order Before Time

CF 000 proves that lawful articulation is strictly dependence-ordered before time, causality, geometry, or differentiability are admitted. A later articulation may appear only if the earlier invariant-bearing conditions that make it lawful have already been borne. Same-domain saturation is therefore not a temporal halt but a logical exhaustion of all genuinely new invariant-bearing articulation available within the present admitted class.

Void Debt, Minimum Protection, and Forced Orthogonal Re-Articulation

When same-domain articulation is exhausted while unresolved burden remains, continuation does not simply stop. By A(-1), void debt begins exactly at that saturation limit. Non-discharge forbids collapse; minimum protection forbids importing more structure than is necessary to keep the invariant borne. Therefore continuation must occur by the *minimum admissible orthogonal re-articulation*. This rule is not local to one stage. It is the constitutional generator of all later effective invariants and effective axioms.

The Primitive Seed: 1D $\to$ 2D, i, and π

The primitive origin of the stationary-action law is already present at the first forced orthogonal articulation. CF 000 closes that once 1D same-axis articulation is saturated while discharge remains impossible, a second irreducible axis is forced and 2D is admitted. Algebra is first licensed only there. The orthogonal rotation structure (ORS) acts discretely by

$$R(1,0) = (0,1), \qquad R(0,1) = (-1,0),$$

so the atomic orthogonal articulation step is the quarter-turn re-expression borne algebraically by i, with $i^2 = -1$. CF 00 then closes the continuous ORS orbit and proves that π is the unique least positive half-turn attainment: the minimal completion/opening condition at which the opposite pole is reached in continuous rotational expression. Thus:

- i is the first algebraic witness of the atomic orthogonal articulation step;
- π is the least completion/opening measure of accumulated orthogonal articulation.

Stationary action is therefore not generic minimisation. It is the minimum admissible accumulation of orthogonal articulation required to reach the relevant opening condition without avoidable first-order excess.

Temporal Admission as a New Effective-Invariant Domain

Once temporal relation becomes admissible, articulation histories become paths between endpoint states. The primitive law does not disappear there. It becomes a new domain-proper effective invariant on path space. In that domain, minimum protection says: among admissible endpoint-fixed continuations, the branch will not pay more accumulated orthogonal articulation cost than is required to keep the invariant borne without discharge at first order. This temporal-path effective invariant is the principle of stationary action, historically known as the principle of least action.

3 Formal Concepts on the Temporal Path Domain

Admissible Path

Definition 1 (Admissible path)**.** An *admissible path* γ between fixed endpoint configurations q_i and q_f is a map

$$\gamma : [t_i, t_f] \to \mathcal{Q},$$

where $\mathcal{Q}$ is the space of admissible configurations, such that for every $t \in [t_i, t_f]$, the state $\gamma(t)$ lawfully bears the inherited invariant stack and does not discharge. The endpoints satisfy

$$\gamma(t_i) = q_i, \qquad \gamma(t_f) = q_f.$$

Endpoint-Fixed Admissible Variation

Definition 2 (Endpoint-fixed admissible variation)**.** An *endpoint-fixed admissible variation* of a path γ is a one-parameter family γ_ε such that each γ_ε is an admissible path, $\gamma_\varepsilon(t_i) = q_i$, $\gamma_\varepsilon(t_f) = q_f$ for all sufficiently small ε, and $\gamma_0 = \gamma$.

Articulation-Cost Density

Definition 3 (Articulation-cost density)**.** The *articulation-cost density* $L_{\mathcal{I}}(q, \dot{q}, t)$ assigns to each admissible state q and path-velocity $\dot{q}$ at time t the local cost of bearing the invariant along the path. At the primitive level, $L_{\mathcal{I}}$ is not assumed to take any specific realised form such as $T - V$. Rather, it measures the local rate of accumulated burden-bearing orthogonal articulation on the admitted temporal domain.

Variational Regularity

Assumption 1 (Variational regularity)**.** For the variational results of this paper, $L_{\mathcal{I}}(q, \dot{q}, t)$ is assumed regular enough for the first variation of the action to exist and for the integration-by-parts step in the Euler–Lagrange derivation to be valid; in particular, $L_{\mathcal{I}}$ may be taken as C^2 in $(q, \dot{q}, t)$ on the domain of admissible paths. This is the minimum regularity required for the present derivation. In realised later domains, the appropriate regularity is inherited from the lawful-variation and carrier structure already earned in CF 00.

Total Articulation Cost

Definition 4 (Total articulation cost)**.** Given an admissible path γ, its *total articulation cost* (or action) is

$$S_{\mathcal{I}}[\gamma] := \int_{t_i}^{t_f} L_{\mathcal{I}}(q, \dot{q}, t)\, dt. \tag{1}$$

Temporal Effective-Invariant Rule

Definition 5 (Temporal effective-invariant rule)**.** On the admitted temporal path domain, the effective invariant is that continuation between fixed endpoint states proceeds only by those admissible paths whose accumulated articulation cost carries no avoidable first-order excess relative to neighbouring endpoint-fixed admissible paths.

Stationary-Action Selection Principle

Definition 6 (Stationary-action selection principle)**.** Among admissible endpoint-fixed paths, VDM selects only those paths γ for which the first-order variation of the total articulation cost vanishes:

$$\delta S_{\mathcal{I}}[\gamma] = 0. \tag{2}$$

Equivalently, the selected path carries no avoidable first-order excess in accumulated orthogonal articulation cost. This is a stationarity condition, not a claim of global minimization.

4 Primitive Stationary-Action Theorem

Temporal Re-Expression of the Orthogonal Articulation Rule

Before time is admitted, the primitive law already forces continuation by minimum admissible orthogonal re-articulation once same-domain articulation is saturated and discharge is forbidden. When temporal relation is admitted, that same rule is re-expressed on path space. The question is no longer merely which new irreducible axis is forced, but which admissible endpoint-fixed path lawfully bears the invariant with no avoidable first-order excess in accumulated orthogonal articulation burden.

Primitive Stationarity Theorem

Theorem 1 (Primitive stationarity)**.** *Let γ be an admissible path between fixed endpoints q_i and q_f. Then γ is selected by the temporal effective-invariant rule if and only if*

$$\delta S_{\mathcal{I}}[\gamma] = 0 \tag{3}$$

for all endpoint-fixed admissible variations. Equivalently, the selected path is the one that carries no avoidable first-order excess in accumulated invariant-bearing orthogonal articulation cost.

Proof. The selection principle on the temporal path domain requires absence of avoidable *first-order* excess, not global minimization. A saddle point is therefore admissible: if $\delta S_{\mathcal{I}} = 0$, the path is selected at first order even if second-order variations later distinguish minima, maxima, or saddles.

Suppose γ is selected and there exists an endpoint-fixed admissible variation γ_ε such that

$$\frac{d}{d\varepsilon} S_{\mathcal{I}}[\gamma_\varepsilon]\bigg|_{\varepsilon=0} \neq 0.$$

If the derivative is positive in one direction, then reversing the variation produces a negative first-order change. In that case γ would carry avoidable *first-order* excess articulation cost relative to a neighbouring admissible path with the same endpoints, contradicting the temporal effective-invariant rule. If the derivative is negative, γ would again fail the same first-order selection rule. Therefore the only lawful possibility is

$$\delta S_{\mathcal{I}}[\gamma] = 0.$$

Conversely, if $\delta S_{\mathcal{I}}[\gamma] = 0$, then no neighbouring endpoint-fixed admissible path reduces the total articulation cost to first order. Hence γ carries no avoidable first-order excess burden and is selected by the temporal effective-invariant rule. This is exactly the path-domain expression of the same law that, at the primitive level, forced minimum admissible orthogonal re-articulation under non-discharge. □

Euler–Lagrange Equations

Theorem 2 (Euler–Lagrange equations)**.** *Let $q(t)$ be an admissible path satisfying Equation* (3)*. Under Assumption 1, the following local equation holds:*

$$\frac{d}{dt}\frac{\partial L_{\mathcal{I}}}{\partial \dot{q}} - \frac{\partial L_{\mathcal{I}}}{\partial q} = 0. \tag{4}$$

This is the local path-level non-discharge balance law: the articulation cost cannot be reduced by infinitesimal admissible deformations of the selected path.

Proof. Consider an endpoint-fixed admissible variation

$$q_\varepsilon(t) = q(t) + \varepsilon\eta(t), \qquad \eta(t_i) = \eta(t_f) = 0.$$

By Assumption 1, the first variation exists and may be expanded to first order:

$$\delta S_{\mathcal{I}} = \int_{t_i}^{t_f} \left(\frac{\partial L_{\mathcal{I}}}{\partial q}\eta + \frac{\partial L_{\mathcal{I}}}{\partial \dot{q}}\dot{\eta} \right) dt.$$

Integrating the second term by parts and using the endpoint conditions gives

$$\delta S_{\mathcal{I}} = \int_{t_i}^{t_f} \left(\frac{\partial L_{\mathcal{I}}}{\partial q} - \frac{d}{dt}\frac{\partial L_{\mathcal{I}}}{\partial \dot{q}} \right) \eta \, dt.$$

Because $\eta(t)$ is arbitrary among admissible endpoint-fixed variations, stationarity requires

$$\frac{d}{dt}\frac{\partial L_{\mathcal{I}}}{\partial \dot{q}} - \frac{\partial L_{\mathcal{I}}}{\partial q} = 0.$$

□

Remark 1 (Path-level two-pole balance and CF01). The two terms in Equation (4) express the two poles of the primitive invariant at the path level. The term

$$\frac{d}{dt}\frac{\partial L_{\mathcal{I}}}{\partial \dot{q}}$$

is the propagation-resistance limb: it measures resistance to deformation of the path's forward articulation. The term

$$\frac{\partial L_{\mathcal{I}}}{\partial q}$$

is the configuration-burden limb: it measures the local burden gradient of configuration change. Their equality at each point on the selected path is the local non-discharge condition. In this sense, CF 14 gives the variational path-space form of the same invariant balance that CF 01 makes explicit in metriplectic language as the J- and M-limb split. Least action and the metriplectic split are therefore not separate organizing laws. They are distinct formal expressions of the same primitive balance, closed at different stages of the series.

5 Realised Variational Principles in Dependency Order

Once the primitive theorem is closed, the known variational principles can be ordered by the additional structure they require. The order below runs from least dependent to most dependent. In each case, the realised variational principle is not a new primitive beginning. It is a domain-specific effective invariant generated from the same root rule.

1. Maupertuis' Principle / Fixed-Energy Conservative Realisation

Inherited dependencies. Primitive non-discharge and dependence order; lawful variation and conservative hosting.
New admitted structure. Conservative motion at fixed total energy, with time not treated as the fixed variational endpoint variable.
Realised variational principle. On a fixed-energy manifold, the conservative path is selected by stationarity of the abbreviated action

$$\delta \int p \cdot dq = 0 \qquad \text{at fixed } E. \tag{5}$$

This is the reduced configuration-space J-limb realisation of the same primitive rule: among admissible fixed-energy trajectories, the selected curve carries no avoidable first-order excess conservative articulation burden.

2. Hamilton's Principle / Classical Mechanics

Inherited dependencies. Primitive non-discharge and dependence order (CF 000, A(-1)); lawful variation, differentiability, carrier structure, and the possibility of continuous path description (CF 00).
New admitted structure. Mechanical sector with propagative and stored-burden decomposition.
Realised variational principle. In the classical mechanical sector, the primitive articulation-cost density realises as

$$L_{\mathcal{I}} \rightsquigarrow T - V, \tag{6}$$

where T is the kinetic/propagation term and V is the potential/stored-burden term. Substituting Equation (6) into the primitive stationarity rule yields Hamilton's principle:

$$\delta \int_{t_i}^{t_f} (T - V)\, dt = 0. \tag{7}$$

The Euler–Lagrange equation then becomes

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = 0,$$

that is, the classical equation of motion. This is the first full fixed-endpoint realised descendant of the primitive stationary-action rule.

3. Thermodynamic-Potential Extremisation

Inherited dependencies. Primitive stationary-action rule plus the thermodynamic/contact/metriplectic sector.
New admitted structure. Equilibrium constraint surfaces, thermodynamic conjugacy, and Legendre-transformed state functions.
Realised variational principle. At equilibrium, thermodynamic admissibility means the system may vary among states consistent with the chosen external constraints. The relevant realised articulation-cost is therefore a thermodynamic potential whose first variation vanishes on the appropriate constraint surface. Examples include:

$$\delta U = 0 \qquad \text{at fixed } (S, V, \ldots), \tag{8}$$

$$\delta F = 0 \qquad \text{at fixed } (T, V, \ldots), \tag{9}$$

$$\delta H = 0 \qquad \text{at fixed } (S, P, \ldots), \tag{10}$$

$$\delta G = 0 \qquad \text{at fixed } (T, P, \ldots). \tag{11}$$

Each thermodynamic potential is a Legendre transform of the internal energy under different constraint choices, and the stationarity condition $\delta\Phi = 0$ for each potential is the same primitive minimum-protection rule expressed on the appropriate thermodynamic constraint surface. CF 02 carries the contact-geometric unification of these Legendre transforms.

4. Onsager's Variational Principle / Dissipative Realisation

Inherited dependencies. Primitive two-pole non-discharge; thermodynamic/contact structure; metriplectic reversible/irreversible split.
New admitted structure. Irreversible near-equilibrium flow, force–flux relations, and entropy-production hosting.
Realised variational principle. CF 02 has already established the contact-geometric and metriplectic hosting of irreversible evolution, including the force–flux decomposition of the M-limb. CF 14 inherits that structure and closes the M-limb variational gate here.

Let X_i denote the thermodynamic forces and J_i the conjugate irreversible fluxes established in CF 02. The Onsager dissipation functional is

$$\Phi(J) := \tfrac{1}{2} \sum_{i,j} L_{ij}\, J_i J_j, \tag{12}$$

where L_{ij} are the Onsager coefficients satisfying $L_{ij} = L_{ji}$ (Onsager reciprocity, inherited from the time-reversal symmetry of the M-limb). The total entropy-production rate along an admissible irreversible trajectory is

$$\dot{\Sigma} = \sum_i X_i J_i. \tag{13}$$

Onsager's variational principle selects the actual irreversible trajectory by stationarity of $\dot{\Sigma} - \Phi$ with respect to variations of the fluxes J_i at fixed thermodynamic forces X_i:

$$\frac{\partial}{\partial J_k}\big(\dot{\Sigma} - \Phi\big) = 0 \quad\Longrightarrow\quad X_k = \sum_j L_{kj}\, J_j. \tag{14}$$

These are the linear irreversible constitutive relations: the actual fluxes are the unique stationary point of the dissipation functional at given forces. In VDM terms, this is the M-limb path-space selection law: among admissible dissipative trajectories consistent with the thermodynamic forces, the selected trajectory bears no avoidable first-order excess dissipative burden. The J-limb selects conservative stationarity ($\delta S_{\mathcal{I}} = 0$); the M-limb selects dissipative stationarity ($\partial(\dot{\Sigma} - \Phi)/\partial J_k = 0$). Together they close the full metriplectic variational structure of CF 14.

5. Generic Local Field Action

Inherited dependencies. Primitive stationary-action theorem, lawful variation, differentiable carrier structure, and field-supporting domain.
New admitted structure. A field ϕ with local density depending on spacetime derivatives.
Realised variational principle. The general local field-theoretic action takes the form

$$S[\phi] = \int \mathcal{L}(\phi, \partial_\mu \phi, x)\, d^d x. \tag{15}$$

Applying the same endpoint-/boundary-fixed variational logic yields the field Euler–Lagrange equation

$$\partial_\mu\left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\right) - \frac{\partial \mathcal{L}}{\partial \phi} = 0. \tag{16}$$

This is the generic field-space realisation of the same primitive balance law.

6. Maxwell / Gauge Field Realisation

Inherited dependencies. Generic field-action stationarity, gauge redundancy, Berry/gauge structure, and the gauge sector closed later in the series.
New admitted structure. Electromagnetic potential A_μ and curvature $F_{\mu\nu}$.
Realised variational principle. For the electromagnetic field,

$$S_{\text{Maxwell}}[A] = -\frac{1}{4}\int F_{\mu\nu} F^{\mu\nu}\, d^4 x. \tag{17}$$

Applying Equation (16) to this realised density yields Maxwell's equations. Hence gauge-field stationarity is not a separate variational miracle; it is the gauge-sector effective invariant generated by the same primitive minimum-saturation rule once gauge-bearing structure is admitted.

7. Einstein–Hilbert and Geodesic Realisation

Inherited dependencies. Primitive stationary-action rule, field-action stationarity, induced geometry/curvature structure, and the gravitational sector.
New admitted structure. Metric $g_{\mu\nu}$, Ricci scalar R, and spacetime volume element.
Realised variational principle. In general relativity, geometry itself bears invariant burden. The gravitational articulation-cost functional realises as

$$S_{\mathrm{EH}}[g] = \frac{1}{16\pi G} \int R\sqrt{-g}\, d^4x. \tag{18}$$

Stationarity of S_{EH} yields the Einstein field equations. Likewise, the geodesic principle is the minimum-protection rule realised on free-fall worldlines: among admissible nearby histories, the selected path bears no avoidable first-order excess spacetime articulation burden. Gravitational variational physics is therefore another effective invariant of the same root law.

8. Schwinger's Quantum Action Principle

Inherited dependencies. Primitive stationary-action law, quantum phase hosting, and operator-level quantum evolution.
New admitted structure. Transition amplitudes between quantum states and operator-valued action variation.
Realised variational principle. At the operator level, the quantum descendant of stationary action is expressed by Schwinger's principle:

$$\delta\langle q_f, t_f \mid q_i, t_i\rangle = \frac{i}{\hbar}\, \langle q_f, t_f \mid \delta S \mid q_i, t_i\rangle. \tag{19}$$

This is more fundamental than any one path representation: it states that quantum transition amplitudes respond to variation of the action by the same ORS phase-weighting rule. The operator equations of motion are generated directly from this principle. In VDM terms, Schwinger's principle is the quantum stationary-action law before committing to any particular path-sum or state-space reduction.

9. Feynman Path Integral / Quantum Path Realisation

Inherited dependencies. Primitive stationary-action theorem; the ORS quarter-turn operator i and half-turn completion/opening condition π from the CF000→CF00 spine; the quantum phase logic carried later in the series.
New admitted structure. Quantum path summation and phase weighting by action.
Realised variational principle. In the quantum realization, the path factor

$$e^{iS/\hbar}$$

is not merely a formal phase weight. The factor i is the atomic orthogonal articulation operator already earned at the first 2D ORS step, while the action S measures the accumulated burden-bearing articulation along the path. The quantity $\hbar$ sets the elementary action scale: the minimum articulation quantum by which accumulated orthogonal articulation enters the ORS phase. The propagator is therefore

$$\langle q_f \mid e^{-iHt/\hbar} \mid q_i\rangle = \int \mathcal{D}[q]\, e^{\frac{i}{\hbar}S[q]}. \tag{20}$$

Stationary phase then has a direct primitive meaning: the selected path is the one whose accumulated orthogonal articulation reaches the required boundary/opening condition with no avoidable first-order excess. As $\hbar \to 0$, neighbouring non-stationary paths acquire rapidly varying ORS phases and cancel, leaving the minimum-saturation path dominant. The path integral is therefore the path-space quantum descendant of the same law already seeded at the first forced orthogonal articulation.

10. Variational Quantum State Principles

Inherited dependencies. Quantum stationary-action law, Hilbert-space state hosting, and operator dynamics.
New admitted structure. Restriction to trial-state manifolds or normalized state families.
Realised variational principle. Several familiar state-space quantum principles are reduced descendants of the same quantum stationary-action law:

- **Rayleigh–Ritz:**
$$\delta\langle\psi \mid H \mid \psi\rangle = 0 \qquad \text{subject to } \langle\psi \mid \psi\rangle = 1, \tag{21}$$
selecting stationary energy expectation states, especially ground-state approximants.
- **Time-dependent variational principles** (e.g. McLachlan / Dirac–Frenkel): evolution on a restricted trial manifold is selected by stationarity of the projected dynamical residual, for example
$$\delta \left\| \left(i\hbar \frac{d}{dt} - H \right) \psi(\lambda(t)) \right\| = 0. \tag{22}$$

These are not independent quantum postulates. They are state-manifold reductions of the same stationary-action law already expressed more generally by Schwinger and by the path integral.

6 Forward Predictions and Program Map

The temporal effective-invariant law yields a clear forward program.

P1. **Universal path-level stationarity.** Any domain that lawfully admits temporal relation must inherit an endpoint-/boundary-fixed stationarity principle expressing the absence of avoidable first-order excess articulation cost.

P2. **Orthogonal-origin continuity.** Realised variational principles must remain traceable to the primitive orthogonal articulation rule already closed through the 1D $\to$ 2D anti-discharge transition.

P3. **Local non-discharge balance.** Every realised variational law must admit a local balance equation analogous to Euler–Lagrange, expressing propagation-resistance versus configuration-burden.

P4. **Metriplectic compatibility.** In domains where the explicit $J \oplus M$ split has been earned, the variational law and the metriplectic balance law must agree as different expressions of the same invariant balance.

P5. **Dissipative variational hosting.** The M-limb must admit its own path-space variational law, realised near equilibrium as an Onsager-type dissipative principle and globally hosted by the CF 02 contact/metriplectic framework.

P6. **Thermodynamic Legendre completeness.** The principal thermodynamic potentials must appear as constraint-surface variants of the same root variational law, not as unrelated extremal rules.

P7. **Quantum operator and path completeness.** Schwinger's principle, path-integral stationary phase, and state-space variational descendants must all be recoverable as compatible quantum realisations of the same stationary-action law.

7 Falsification Criteria

CF 14 is falsified if any of the following occur:

F1. Orthogonal-origin failure. Stationary action can be shown to be independent of the primitive orthogonal anti-discharge mechanism already closed in the CF000$\to$CF00 spine.

F2. Temporal effective-invariant failure. A domain lawfully admits temporal relation, but the selected admissible path carries avoidable first-order excess articulation cost.

F3. Non-stationary selected path. A physically selected path between fixed endpoints satisfies $\delta S \neq 0$ under admissible endpoint-fixed variations.

F4. Local imbalance. The derived local law fails to take Euler–Lagrange form, so the propagation-resistance and configuration-burden limbs do not balance.

F5. Metriplectic inconsistency. In a domain where the explicit $J \oplus M$ split has already been earned, the variational law and the metriplectic law cannot be reconciled as expressions of the same primitive invariant balance.

F6. Dissipative variational failure. The M-limb admits dissipative hosting but no corresponding variational selection law for irreversible trajectories.

F7. Quantum variational incompleteness. Schwinger's operator principle, path-integral stationary phase, and state-space descendants cannot be reconciled as realizations of one inherited quantum stationary-action rule.

F8. Failure of realised descendants. Any of the realised principles listed in Section 5 admits a lawful counterexample that violates the inherited stationarity law while preserving the same dependency stack.

8 Validation Gates

Analytical Gates

Gate: G1: Effective-invariant constitutional bridge *Threshold: Complete.*

Show that stationary action is not an independent axiom but the temporal-path effective invariant generated by the primitive bifurcation law under non-discharge, void debt, and minimum protection.
Status: Proved in this paper.

Gate: G2: Orthogonal-origin bridge via i and π *Threshold: Complete.*

Demonstrate that the first forced orthogonal articulation already contains the primitive seed of stationary action: i as atomic orthogonal articulation step and π as least completion/opening condition.
Status: Stated and integrated in this paper from the CF000→CF00 spine.

Gate: G3: Primitive stationarity theorem *Threshold: Complete.*

Prove that the temporal effective-invariant rule on endpoint-fixed admissible paths is exactly $\delta S_{\mathcal{I}} = 0$.
Status: Proved in Theorem 1.

Gate: G4: Euler–Lagrange derivation *Threshold: Complete.*

Derive the Euler–Lagrange equation from stationarity and interpret it as the local path-level non-discharge law.
Status: Proved in Theorem 2.

Gate: G5: Variational / metriplectic identification *Threshold: Integrated.*

Show that the Euler–Lagrange balance derived here is the path-space expression of the same two-pole balance later written explicitly as the $J \oplus M$ split in CF 01.
Status: Integrated interpretively in Remark 1; explicit QGT-to-metriplectic construction belongs to CF 01.

Realisation Gates

Gate: G6: Conservative realisations *Threshold: Closure.*

Show that Maupertuis' principle and Hamilton's principle are conservative J-limb descendants of the primitive stationary-action law under their respective endpoint/constraint conditions.
Status: Derived schematically here; formal closure belongs to the mechanical sector in the canon.

Gate: G7: Thermodynamic Legendre realisation *Threshold: Closure.*

Show that the principal thermodynamic potentials arise as constraint-surface realizations of the same root law.
Status: Closed in this paper, inheriting CF 02. The contact-geometric unification of the Legendre transforms is already established in CF 02; CF 14 inherits that structure and closes the constraint-surface stationarity conditions for all four principal thermodynamic potentials here.

Gate: G8: Dissipative M-limb variational realisation *Threshold: Closed.*

Show that irreversible trajectories admit an Onsager-type variational selection law consistent with the M-limb and with the CF 02 contact/metriplectic framework. CF 02 has already established the force–flux structure of irreversible evolution. CF 14 inherits that structure and closes the gate: stationarity of $\dot{\Sigma} - \Phi$ with respect to irreversible fluxes at fixed forces yields the linear constitutive relations as the M-limb path-selection law (Equation (14)).
Status: Closed in this paper, inheriting CF 02.

Gate: G9: Field, gauge, and gravitational realisations *Threshold: Closure.*

Show that local field stationarity, Maxwell action stationarity, and Einstein–Hilbert stationarity are realised descendants of the same law.
Status: Derived schematically here; formal closure belongs to the relevant later CFs.

Gate: G10: Quantum operator / path / state completeness *Threshold: Closure.*

Show that Schwinger's principle, the path integral, and state-space variational descendants are compatible realizations of one inherited quantum stationary-action law.
Status: Derived conceptually here from the ORS/action bridge; fuller closure belongs to the dedicated quantum CF line.

9 Discussion

Why the Result Is Foundational

Stationary action is often taught as a postulate summarising broad empirical regularities. CF 14 shows that it is more basic and more constrained than that. It is a domain-proper effective invariant generated by one primitive law. Because it arises from the same non-discharge/minimum-protection rule that already forced the first orthogonal articulation, it belongs at the foundational level of the series rather than as a late optional formal convenience.

Why i and π Matter Here

The orthogonal articulation story is not decorative. It is the concrete origin of the stationary-action law. The quarter-turn operator i is the first algebraic witness of the atomic orthogonal articulation step; π is the first completion/opening measure of continuous opposite-pole attainment. CF 14 matters because it shows that the later action law is not a different optimization principle layered on top of that machinery. It is the temporal-path re-expression of that same machinery.

Stationary Action and the Effective-Invariant Rule

The constitutional lesson of A(-1) is that later invariants are effective, cumulative, and minimally stabilised. CF 14 makes stationary action one of those effective invariants. Once time is admitted, the primitive law generates a new path-domain rule: among admissible endpoint-fixed continuations, the selected path is the one that bears no avoidable first-order excess accumulated orthogonal articulation cost. This is why the stationary-action law can work across scales. It is not tied to one later sector. It is the temporal effective-invariant rule itself.

Why the M-Limb Must Also Appear Variationally

Because VDM is metriplectic rather than purely conservative, a complete variational bridge cannot stop at Hamiltonian/J-limb stationarity. The dissipative limb must also admit a path-space selection principle. CF 14 therefore explicitly includes Onsager's principle as the realised near-equilibrium M-limb variational descendant, even though the full thermodynamic/contact hosting and GENERIC-style geometry close in CF 02. Without that closure the series would derive the conservative variational law while only implying the dissipative one. Because CF 02 is already closed upstream, CF 14 can and must inherit its force–flux structure and close the M-limb variational gate here rather than deferring it.

Connection to CF01, CF02, and Later Physics

CF 14 closes a crucial architectural loop in the series. CF 01 already gives the explicit QGT-to-metriplectic $J \oplus M$ split. CF 02 gives the contact-geometric and thermodynamic hosting of irreversible evolution. CF 14 shows why variational laws and two-limb evolution laws should exist at all: both are descendants of the same primitive invariant balance. The later realized principles therefore do not float independently. Conservative mechanics, thermodynamic extremisation, dissipative Onsager flow, gauge theory, gravity, quantum operator variation, path summation, and state-space variational methods are all dependency-ordered closures of the same root law.

Open Questions

- **Higher-order stationarity.** What is the correct primitive interpretation of second and higher variations in non-convex domains?
- **Boundary and noncompact effects.** How do domain boundaries, noncompact supports, and boundary-hosting modify the path-level effective invariant?
- **Global dissipative variational law.** What is the strongest non-near-equilibrium M-limb variational principle compatible with the full metriplectic/contact structure?
- **Exotic realised forms.** Which later sectors require new effective axioms beyond the standard variational descendants discussed here?

10 Conclusions

CF 14 derives the principle of stationary action from the Primitive Bifurcation Law by placing it in its correct constitutional role. There is one primitive law; later invariants and later axioms are effective rather than primitive; and the first forced orthogonal articulation already contains the seed of stationary action through the quarter-turn operator i and the half-turn completion/opening measure π. Once temporal relation is admitted, the same law generates a temporal-path effective invariant: among admissible endpoint-fixed histories, continuation proceeds only by paths carrying no avoidable first-order excess in accumulated orthogonal articulation cost. This yields the stationarity condition $\delta S_{\mathcal{I}} = 0$, from which the Euler–Lagrange equation follows as the local path-level non-discharge balance law. The known variational principles can then be ordered by dependency burden and derived as realised descendants of that same rule, including conservative, thermodynamic, dissipative, field-theoretic, gravitational, operator-quantum, path-quantum, and state-space quantum realizations. Stationary action is therefore not a separate postulate and not generic minimisation. It is the temporal effective invariant of minimum admissible orthogonal articulation.

References

[1] J. K. Lietz. A(-1): Universal Axiom of Primitive Bifurcation and Orthogonal Articulation in VDM. *Neuroca, Inc.*, March 2026.

[2] J. K. Lietz. CF000: Complete Formalism — Primitive Distinguishability and Orthogonal Articulation. *Neuroca, Inc.*, March 2026.

[3] J. K. Lietz. CF00: Complete Formalism — Induced Geometry and Emergent Dynamics. *Neuroca, Inc.*, March 2026.

[4] J. K. Lietz. CF01: Complete Formalism — Quantum Geometric Tensor to Metriplectic Brackets in VDM. *Neuroca, Inc.*, March 2026.

[5] J. K. Lietz. CF02: Complete Formalism — Contact Geometry to Metriplectic Evolution. *Neuroca, Inc.*, 2025–2026.

[6] J. K. Lietz. CF09: Complete Formalism — Gauge Field Emergence via Berry Connection in VDM. *Neuroca, Inc.*, 2025–2026.

[7] J. K. Lietz. CF12: Complete Formalism — Gravity and Black Hole Thermodynamics from VDM Invariant Burden Density. *Neuroca, Inc.*, March 2026.

[8] J. K. Lietz. CF13: Complete Formalism — π, Transcendence, Chirality, and the Nielsen–Ninomiya Obstruction in VDM. *Neuroca, Inc.*, March 2026.

[9] L. Onsager. Reciprocal relations in irreversible processes. *Physical Review*, 37–38, 1931.

[10] J. Schwinger. The theory of quantized fields. I. *Physical Review*, 82, 1951.

[11] J. K. Lietz. The Lietz Infinity Resolution Conjecture: Hierarchical Scale-Breaking in Tachyonic Metriplectic Systems. *Neuroca, Inc.*, October 2025.